Samuel Baumgartner

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Profile

Electrical Engineering student with a GPA of 5.7/6.0 (top 5%), planning an MSc in Robotics. Combines analytical training with leadership, teaching, international study, and rapid multidisciplinary project work, with an interest in how engineering decisions meet public systems, sustainability, and entrepreneurship.

Education

ETH Zurich *BSc Electrical Engineering and Information Technology* *2023–present*

- 135 ECTS completed; focus on control, signal processing, embedded systems, and robotics.
- Selected coursework: Control Systems, Computational Thinking, and Signal and System Theory.
- Broader-context coursework: Swiss Environmental Policy; Soccer Analytics.

Institute of Science Tokyo, Japan *Exchange semester in Tokyo* *Apr–Aug 2026*

- Interdisciplinary coursework including data analytics and machine learning, biomechanics and digital human modelling, optics, and solid dynamics.
- Adapted to a different academic culture while studying Japanese at approximately JLPT N4 level.

Leadership and Experience

ETH Zurich *Teaching Assistant* *2024–present*

- Lead weekly exercise sessions and support students in circuit analysis and magnetic fields.
- Explain complex material to students with different levels of preparation and provide individual feedback.

Swiss Armed Forces *Sergeant, technical function* *2022–2023*

- Led a 25-person unit and held technical responsibility for digital systems in a mission-critical environment.
- Developed discipline, clear communication, and decision-making under pressure.

Selected Multidisciplinary Projects

UrbanLens *NASA Space Apps Challenge, 3rd place* *Oct 2025*

- Built the Python API, analysis pipeline, and public-transport accessibility layer for a healthy-cities platform combining mobility, heat, pollution, vegetation, population, and land-use data.
- Worked in a five-person team to turn heterogeneous evidence into an interactive tool for comparing urban priorities.

Flora AI *StartHack, 2nd of 26 teams* *Mar 2026*

- Built an agent-driven indoor-farming and nutrition simulation combining AI, environmental sensors, embedded hardware, and a web interface in 36 hours.

Guidely *Swiss AI Hack, 2nd of 40 teams* *Nov 2025*

- Developed the core AI guidance pipeline and desktop overlay within 23 hours while teammates focused on user value, business model, and pitch.

Progress Pact *StartHack finalist* *Mar 2025*

- Co-created a workplace well-being platform combining AI coaching, goals, multilingual communication, and social participation.

Skills and Languages

Methods and tools: Python, C++, data analysis, machine learning, control systems, embedded systems, full-stack prototyping, Git, Linux, Docker.
Languages: German (native), English (fluent), French (intermediate, Swiss Matura), Japanese (approximately JLPT N4, self-assessed).
Interests: Robotics, technology policy, sustainable systems, entrepreneurship, beekeeping, and fitness.